

Australian Math Olympiad Day 1 2016 — P1/4

Jonathan Kasongo

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Find all positive integers n such that $2^n + 7^n$ is a perfect square.

Solution

The only n that works is $n = 1$, since $2^1 + 7^1 = 3^2$.

Let $m \in \mathbb{N}$. We are looking for n such that $2^n + 7^n = m^2$. Reducing modulo 3 yields $(-1)^n + 1^n \equiv m^2 \pmod{3}$. But since perfect squares are only 0 or 1 modulo 3, it must be the case that n is odd and $3 \mid m^2$.

Now suppose, for the sake of contradiction, that $n \geq 3$ is a solution. We can write $n = 2k + 1$ for $k \in \mathbb{N}$. Consider the following

$$\begin{aligned} m^2 &\equiv 2^{2k+1} + 7^{2k+1} \equiv 0 \pmod{3} \\ m^4 &\equiv 2^{4k+2} + 7^{4k+2} + 14^{2k+1} \equiv 1 + 1 - 1 \equiv 1 \pmod{3} \end{aligned}$$

but this is a contradiction since $3 \nmid m^4$. Thus indeed only $n = 1$ works. ■